

8.1 Classwork: Rational Functions (HL)

A graphing calculator is allowed. Answer on lined or graph paper; label each part clearly. Show enough algebra that a marker can follow your reasoning — a GDC answer alone is not sufficient unless the part says “write down”.

1. [Maximum mark: 12] The function f is defined by $f(x) = \frac{x^2 - 3x - 4}{x - 2}$, $x \in \mathbb{R}$, $x \neq 2$.

(a) Write down the equation of the vertical asymptote of the graph of f . [1]

(b) Find the coordinates of the points where the graph of f crosses the x -axis, and the y -axis. [3]

(c) Using polynomial division, or otherwise, show that $f(x) = x - 1 - \frac{6}{x - 2}$, and hence state the equation of the oblique asymptote of f . [4]

(d) Using your GDC, sketch the graph of $y = f(x)$ for $-4 \leq x \leq 8$. Label the asymptotes (dashed) and the three intercepts. [2]

(e) Solve the inequality $f(x) > 0$. [2]

2. [Maximum mark: 10] Let $f(x) = \frac{1}{x^2 - 2x - 8}$, $x \in \mathbb{R}$, $x \neq -2$, $x \neq 4$.

(a) State the equations of the two vertical asymptotes and the horizontal asymptote of the graph of f . [3]

(b) Find the y -intercept of f . [1]

(c) Show that f has a single local extremum, and find its coordinates. State whether it is a local maximum or a local minimum, and justify your answer. [4]

(d) Sketch the graph of $y = f(x)$, indicating asymptotes (dashed), the y -intercept, and the local extremum. [2]

3. [Maximum mark: 8] The function f is defined by $f(x) = \frac{x-2}{x^2-9}$, $x \in \mathbb{R}$, $x \neq \pm 3$.
- (a) State the equations of all three asymptotes of the graph of f . [3]
 - (b) Find the coordinates of the x -intercept and the y -intercept of f . [2]
 - (c) Solve the inequality $f(x) \geq 0$. Express your answer using interval notation. [3]
4. [Maximum mark: 6] The function g is defined by $g(x) = \frac{ax+4}{3-x}$, $x \in \mathbb{R}$, $x \neq 3$, $a \in \mathbb{R}$.
- (a) Given that $g(x) = g^{-1}(x)$ for all x in the domain of g , determine the value of a . [4]
 - (b) For this value of a , write down the equation of the line of symmetry of the graph of $y = g(x)$, and explain why every self-inverse function has its graph symmetric about this line. [2]